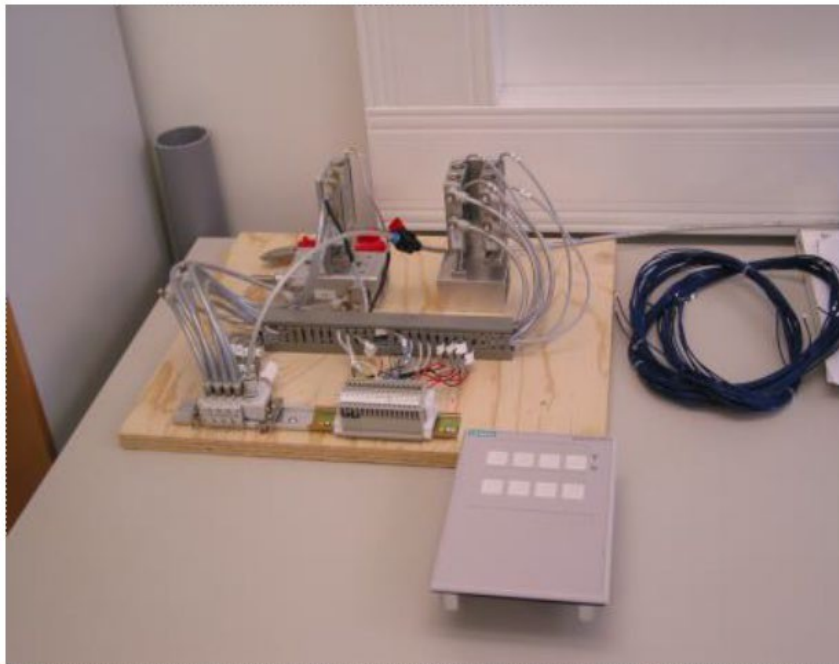

CUTTING APPLICATION**1 INTRODUCTION**

For work, a plywood mounting plate is used, on top of which a "cut-off device" consisting of four cylinders is built. The hardware is pre-connected, but check the connections to begin with the task. Depressurize the equipment before inspection! The task is to do a program in logic that will control the cylinders in accordance with the assignment.



CUTTING APPLICATION**2 INFORMATION ABOUT THE WORKSTATION****Programmable logic**

The programmable logic at the base is a Siemens S7 1214C. It has an integrated Profinet interface, as well as binary inputs and outputs that are used in this work. The I/O connections of logic to the terminal blocks are made in such a way that no changes need to be made to the connections. The top row of terminal blocks are output outputs: 1 and 2 are the voltage inputs and 3-6 are Q0.0-Q0.3. The middle row of terminal blocks is input inputs: 16-23 are I0.0-I1.0. In the lowest row of terminal blocks are 24 V source voltages.

Cylinder position detection

The open/closed positions of the cylinders are identified by compact approach switches attached to the grooves machined into the cylinder. The operation of the proximity switch in this case is based on the presence of magnetic matter in the piston inside the cylinder, which is recognized by the clutch.

Valves

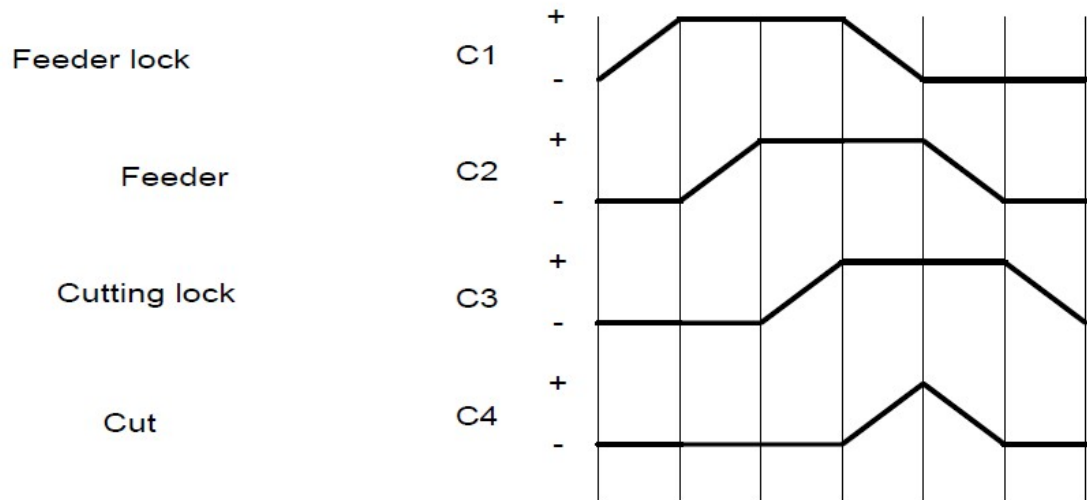
SMC valves are used in the work. The valves are controlled by a voltage of 24VDC (from the outputs), and the function of the valves is to control the passage of compressed air to the cylinders.

CUTTING APPLICATION**3 SAFETY**

At the workstation, normal caution should be exercised when working with electrical and compressed air equipment. N.B! Risk of squeezing!

4 TASK

1. Let's get acquainted with the main features of the equipment. Find out for yourself how the equipment works with this instruction and, if necessary, also ask the instructor.
2. Launch the TIA Portal, create a project and add the correct CPU to it. Test loading HW and an empty program into logic – if there are any problems, work out with the instructor.
3. Make a program that implements the functions described below (test the operation of the program in monitoring mode as necessary to know if you are on the right track):
 - The app starts from the Start pushbutton and stops in an emergency from the Stop pushbutton.
 - At the beginning of the application, there is a withdrawal period of 2 seconds, before the movement of cylinder C1.
 - The operation of the application after the withdrawal period is as follows: the device feeds material to the cutting blade, performs a cut-off and returns to the basic mode. See Figure 2.
 - After the cut-off sequence is complete, the application must return to its initial state (i.e. stop) until it is restarted from the Start push-button again.

CUTTING APPLICATION

After testing the program, present it to the teacher (demonstrate the activity and present the hw configuration, program and symbol/tag you have made).

CUTTING APPLICATION**5 ATTACHMENTS****Annex 1. I/O's addresses**

ADDRESS	EXPLANATION
I1.1	"Start" -push-button input
I1.2	"Stop" -push-button input
I0.0	C1 in position
I0.1	C1 out position
I0.2	C2 in position
I0.3	C2 out position
I0.4	C3 in position
I0.5	C3 out position
I0.6	C4 in position
I0.7	C4 out position
Q0.0	C1 out
Q0.1	C2 out
Q0.2	C3 out
Q0.3	C4 out